Rajiv Choudhury

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Summary

Incoming CS graduate student at **UC Davis** (Fall 2026) and B.Tech graduate from **IIIT Guwahati** with a strong foundation in **deep learning, Python, and C++**. Experienced in optimizing **machine learning** workloads, **GPU computing**, and implementing robust software engineering practices. Eager to leverage a strong foundation in software engineering and algorithmic analysis to mentor undergraduate students in core computer science concepts.

Education

Graduate Student	MS (Computer Science) at **UC Davis**	
Class of 2024	B.Tech (Computer Science Engineering) at **IIIT Guwahati**	(CPI: 9.13/10)
Class of 2020	Class 12th, MBS Public School (CBSE)	(90.80%)
Class of 2018	Class 10th, DAV Public School (CBSE)	(90.60%)

Work Experience

Software Engineer — Samsung R&D Institute India - Delhi Jul 2024 – Aug 2026

- Developed and maintained **Tizen OS** features using **C++** and **C#**; conducted peer code reviews and authored technical documentation to ensure maintainable code standards across the engineering team.
- Developed zero-contact USB presets for **E-paper signage** to improve accessibility, and a certificate-expiration warning for **10,000+ enterprise displays**, earning the team **Best Feature Award**.
- Integrated offline **OpenAI Whisper** speech-to-text on Android Displays, translating complex AI models into accessible tools for end-users.

Research Intern — Centre for Networked Intelligence, IISc Bengaluru Jan 2024 – Jun 2024

- Conducted research on **NVIDIA GPU architecture** and parallel computing, distilling complex HPC concepts into accessible technical summaries.
- Proposed and documented a scheduling method achieving up to **80% improvement** in energy efficiency over standard Linux CPU governors. (Published in **ACM eEnergy** conference 2024 Singapore)
- Collaborated with **IBM Research** on GPU scheduling and matrix multiplication optimization by implementing and benchmarking **CUDA C++** machine learning models against **PyTorch**.

Research Intern — IIIT Guwahati May 2022 - Jul 2022

- Worked on **Brain-Computer Interface (BCI)** systems, conducting comprehensive literature reviews on EEG signal classification.
- Implemented **MLP classifiers and Neural Networks**, communicating technical research findings and methodologies effectively. (Published this work in **IEEE DCOSS-IoT** Conf. Abu Dhabi 2024)

Projects

On-Device Whisper AI Assistant Tech: Kotlin, Jetpack Compose, Whisper

- Integrated offline **Whisper** speech-to-text model on Android platforms, writing clean, modular code to expand AI accessibility on low-latency interactive displays.
- Optimized inference pipelines, ensuring seamless integration and high-quality user experience.

Fashion Recommender Tech: Python, Streamlit, TensorFlow

- Created a structured, well-documented image-based fashion recommendation system using **ResNet** embeddings and KNN similarity search, emphasizing clean **Python** architecture.
- Deployed the application with an accessible web interface to visually recommend similar products.

Skills

Languages	Python, C++, CUDA C++, Java, C#, C
Web & Publishing	Google Colab, Jupyter Book, Web Accessibility Principles
Tools & Platforms	Git, GitHub, Unix/Linux Command Line, Docker, AWS, Android, Tizen
Data & AI Tools	Google Gemini, OpenAI Whisper, NumPy, Pandas, Scikit-learn
Frameworks & DBs	PyTorch, TensorFlow, Jetpack Compose, MySQL, MongoDB